| Faculty of: Engineering and Technology | Department: Computer Science and Engineering<br>Program: B.Tech | AY:2026-27 |
|---|---|---|

# INTERNAL ASSESSMENT (I.A.) TEST: I

| Course Title: Data Structures and Applications | Course Code:25CS305 |
|---|---|
| Date: 15/10/26 | Time: 10:00 to 11:30 AM |
| Semester and Section: III ' C' | Max. Marks:30 |

**Note: Answer any three full questions selecting at least one question from each module**

| Q.No. | | QUESTIONS | Marks | CO's | RBT Levels |
|---|---|---|---|---|---|
| | | **MODULE-I** | | | |
| 1 | a) | List and explain dynamic memory allocation functions with examples. | 6 | CO1 | L2 |
| | b) | Assume A=1, B=2, C=3. Evaluate the following postfix expressions:<br>a. A B + C – B A + C $ -<br>b. A B C + * C B A – + * | 6 | CO2 | L4 |
| | | **OR** | | | |
| 2 | a) | Apply suitable data structure to cover following infix to postfix expression<br>i)(A-B) * (D/E) ii)(a*b)+c | 6 | CO2 | L3 |
| | b) | Define Data structures. List and explain the types of data structures.Give example. | 6 | CO1 | L2 |
| | | **MODULE-II** | | | |
| 3 | a) | What is a Queue. Implement the operation of linear queue. | 6 | CO2 | L3 |
| | b) | Implement insertion operations in a singly linked list at the beginning, at the end and after a specified node with suitable algorithms. | 6 | CO2 | L3 |
| | | **OR** | | | |
| 4 | a) | Apply the concept of a circular queue to develop a C program that performs its basic operations. | 6 | CO2 | L3 |
| | b) | Analyse the insertion and deletion operations of a deque at both the front and rear ends | 6 | CO2 | L4 |
| | | **MODULE-III** | | | |
| 5 | a) | What is doubly linked list .Implement its operations and explain with example. | 6 | CO2 | L3 |
| | | **OR** | | | |
| 6 | a) | Implement the following : i)Insert a node at the begining of circular singly linked list ii) Insert a node at the end of circular singly linked list. | 6 | CO2 | L3 |